

5A-456-400

## Engine Test Certificate

Perkins Engines Company Limited  
Eastfield, Peterborough PE1 5FQ  
Great Britain



Sep 07, 2019

This engine has been tested to its maximum rating in accordance with Perkins defined requirements and has performed satisfactorily.

The electronics on this engine have been preset and when the engine has been run-in, the gross power output will be within +/-3% of the gross rating listed below, provided the exhaust and induction systems, fuel and lubricating oils are in accordance with Perkins defined requirements.

| Engine Type   | Perkins Serial Number | Customer |
|---------------|-----------------------|----------|
| 2206A-E13TAG5 | TGFF5079N09656E       | B0080E   |

| Reference                        | Technical Data      |
|----------------------------------|---------------------|
| Emission Capability              | Unregulated Low SFC |
| Engine Speed                     | 50 Hz @1500RPM      |
| Engine Type                      | 2206A-E13TAG5       |
| Fuel Consumption - Prime Power   | 207 g/kWH           |
| Fuel Consumption - Standby Max   | 203 g/kWH           |
| Gross Engine Power - Prime Power | 324.2 kW            |
| Gross Engine Power - Standby Max | 368.4 kW            |
| Gross Rating                     | 324.2 kW            |
| Net Rating                       | 305 kW              |

This engine was built and tested in accordance with our approved ISO 9001 Quality Procedure and stated Rating, as determined by the ISO 3046 (BS5541) Standard.

Performance certified as satisfactory.

Signed,

For and on behalf of Perkins Engines Company Limited - Large Engines Division.

**MUST BE KEPT WITH ENGINE AT ALL TIMES**

Issue 4

SGR 456-400

| VARADA ENERGIES PVT LTD. ( AN ISO 9001-2015 CERTIFIED COMPANY ) |                          |                             |                            |
|-----------------------------------------------------------------|--------------------------|-----------------------------|----------------------------|
| STANDARD- BS 5514                                               |                          | ISO- 8528                   |                            |
| GENERAL INFORMATION                                             |                          |                             |                            |
| Customer Name                                                   |                          | Customer Order No.          | 2                          |
|                                                                 |                          | Test Cell No.               |                            |
| PRODUCT INFORMATION                                             |                          |                             |                            |
| Model                                                           | VEPL-400-CDP             | Genset Serial No.           | VEPL/2020-21/0015          |
| Engine Serial No.                                               | TGFF 5079N09656E         | Alternator Serial No.       | N20C 114662                |
| Brand Name                                                      | PERKINS                  | Alternator Make.            | STAMFORD                   |
| Fuel Type                                                       | HSD                      | Controller Type             | DEEP SEA                   |
| Governor Type                                                   | ELECTRONIC               | Cooling System Type         | WATER COOLED               |
| TEST CONDITION                                                  |                          |                             |                            |
| Altitude (Meters ASL)                                           | 1000                     | Oil Specifications          | PERKINS MAKE               |
| Ambient Temp. (C)                                               | 33                       | Humidity (%)                | 41                         |
| Air Filter Inlet Temp (C)                                       | 38                       | Coolant Mix                 | PERKINS MAKE               |
| PRODUCT OPTIONS CHECKED                                         |                          |                             |                            |
| (e) Emergency Stop                                              | (e) Panel Hooter         | (e) Low Coolant Warning     | (o) High Fuel S/Down       |
| (e) Over Load Protection                                        | (e) High Water Temp      | (o) LET Warning             | (o) High Fuel Warning      |
| (e) Over Speed Protection                                       | (e) Low Lub Oil Pressure | (o) Low Fuel S/Down         | (e) Parallel Bus PT Module |
| (e) Under Voltage Protection                                    | (e) Low Coolant S/Down   | (e) Low Fuel Warning        | (o) Earth Fault S/Down     |
| TECHNICAL DATA (Declared)                                       |                          |                             |                            |
| Service Rating                                                  | 400 kVA                  | Power Factor                | 0.8                        |
| Power Output (kW)                                               | 320                      | Rated Output (kW)           | 320                        |
| Rated Voltage (V)                                               | 400                      | Rated Current (A)           | 556                        |
| Frequency (Hz)                                                  | 60                       | Speed (RPM)                 | 1800                       |
| Number of Phases                                                | 3                        | Performance Class           | H                          |
| TECHNICAL DATA (Measured)                                       |                          |                             |                            |
| Results (No Load)                                               |                          |                             |                            |
| Voltage                                                         | 1 - 2 : 401 Volts        | 2 - 3 : 403 Volts           | 1 - 3 : 403 Volts          |
| No Load Frequency                                               | 60.1                     | No Load Coolant Temp. (C)   | 76                         |
| No Load Lube Press                                              | 3.89 Bar                 |                             |                            |
| Result (Full Load)                                              |                          |                             |                            |
| Voltage                                                         | 1 - 2 : 401 Volts        | 2 - 3 : 400 Volts           | 1 - 3 : 402 Volts          |
| Current                                                         | 1 : 556 Amps             | 2 : 557 Amps                | 3 : 556 Amps               |
| Full Load Frequency                                             | 60 Hz.                   | Full Load Coolant Temp. (C) | 93                         |
| Full Load Lube Press                                            | 3.87 Bar                 | Power Factor                | 1.0                        |
| Prime set tested to 110% of Full Load (if required)             |                          |                             |                            |
|                                                                 |                          | NA                          |                            |
|                                                                 |                          | Date                        | 06.08.2020                 |
|                                                                 |                          | Run Hours                   | 60 mins                    |
|                                                                 |                          | Signature                   |                            |
|                                                                 |                          | Stamp                       |                            |

